

Name: Maths Class:

SYDNEY TECHNICAL HIGH SCHOOL



YEAR 12 HSC COURSE

Extension 1 Mathematics

HSC Task 2

March 2009

TIME ALLOWED: 70 minutes

Instructions:

- Write your name and class at the top of this page, and on all your answer sheets.
- Hand in your answers attached to the rear of this question sheet.
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Marks indicated are a guide only and may be varied at the time of marking

(FOR MARKERS USE ONLY)

1	2	3	4	5	6	TOTAL
/11	/11	/11	/11	/11	/11	/66

QUESTION 1 (11 Marks):

	Marks
(a) (i) Find the points of intersection of the curves $y = x^2$ and $y = 3x - 2$	1
(ii) Calculate the area between the two curves	2
(b) Given that $\frac{d^2y}{dx^2} = 12x - 2$, and that when $x = 0$, $\frac{dy}{dx} = 0$ and $y = 4$, find y in terms of x .	2
(c) The section of the curve $y = \sqrt{4 - x^2}$ between $x=1$ and $x=2$ is rotated about the x -axis.	
(i) describe <u>or draw a sketch of</u> the solid so formed	1
(ii) Find the volume of the solid, leaving your answer in terms of π .	3
(d) By using the substitution $u = 1 - x^2$, or otherwise, find $\int x\sqrt{1 - x^2} \, dx$	2

QUESTION 2 (11 Marks):

(a) Show that if $y = (1 - x)(x + 1)^3$ then $\frac{dy}{dx} = 2(x + 1)^2(1 - 2x)$	2
(b) You are given that $\frac{d^2y}{dx^2} = -12x(x + 1)$. Find all stationary points on the curve and their nature.	4
(c) Find all points of inflexion.	2
(d) Sketch the curve, showing all major features	3

QUESTION 3 (11 marks):

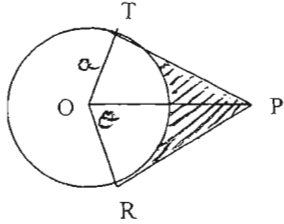
- (a) Find Marks
- (i) $\int \frac{x^3+1}{x^2} dx$ 2
- (ii) $\int_0^4 \frac{dx}{\sqrt{x}}$ 2
- (b) A piece of wire 28 cm long is cut and then bent to form a rectangle and a square.
- (i) If the width of the rectangle is x cm and the length is 3 times its width, show that the sum of the areas of the rectangle and square is given by 2
- $$A = 7x^2 - 28x + 49$$
- (ii) If A is to be a minimum, find the area of the square. 3
(Justify that your answer is a minimum.)
- (c) Differentiate $\sqrt{1-2x}$ and hence, or otherwise, find $\int \frac{1}{\sqrt{1-2x}} dx$ 2

QUESTION 4 (11 Marks):

- (a) The table of values below describes the function $y = f(x)$ 2
- | | | | | | |
|--------|---|-------|-------|-------|-------|
| x | 0 | 0.5 | 1 | 1.5 | 2 |
| $f(x)$ | 1 | 1.649 | 2.718 | 4.482 | 7.389 |
- Using Simpson's Rule with 5 function values, approximate $\int f(x)dx$
Give your answer correct to 2 decimal places.
- (b) Solve $2\sin 2x = \sqrt{3}$ for $0 \leq x \leq \pi$ 4
- (c) Using the substitution $u=1-x$, or otherwise, evaluate 5

$$\int_0^1 x\sqrt{1-x} dx$$

QUESTION 5 (11 Marks):

	Marks
(a) (i) Express $\cos\theta - \sin\theta$ in the form $A\cos(\theta+\alpha)$ where $A>0$ and $0 \leq \alpha \leq \frac{\pi}{2}$	2
(ii) Hence, or otherwise, solve $\cos\theta - \sin\theta = 1 \quad \text{for } 0 \leq \theta \leq 2\pi$	2
(iii) If the domain of part (ii) above was changed to $-\pi \leq \theta \leq \pi$ give the new solutions to $\cos\theta - \sin\theta = 1$	1
(b) Two tangents are drawn to a circle, centre O, of radius a units from a point P. P is $2a$ units from the centre of the circle. THE DIAGRAM IS NOT TO SCALE	
	
(i) Find the area of $\triangle OTP$	2
(ii) Find the size of the angle TOR	1
(iii) Find the area of the shaded section.	3

QUESTION 6 (11 Marks):

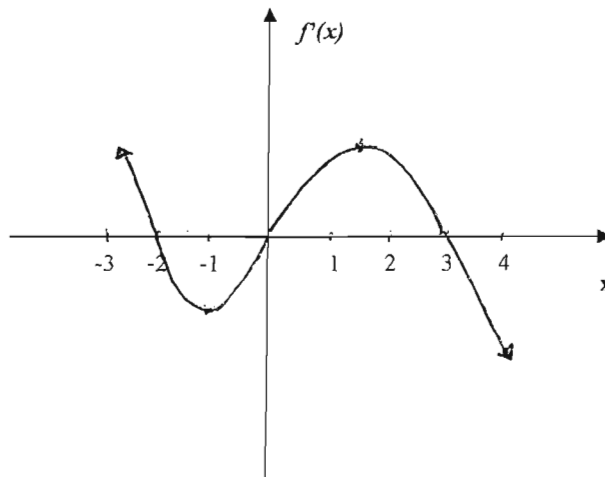
Marks

- (a) For a certain curve $y = f(x)$, the graph of $y = f'(x)$ is sketched below.

4

You are also given that $f(3)=3$, $f(0) = -3$ and $f(-2) = 1$.

Sketch a possible graph of $y=f(x)$ over $-3 \leq x \leq 4$

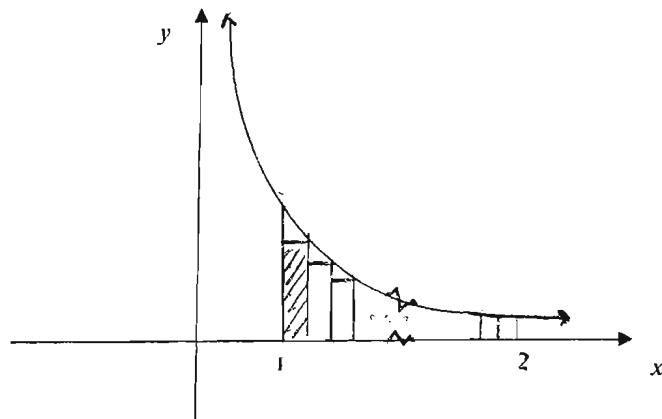


QUESTION 6 continues over the page.....

QUESTION 6 continued.....

Marks

- (b) The curve $y = \frac{1}{x^2}$ is shown below



The area below the curve and between the ordinates $x=1$ and $x=2$ has been divided into n rectangles of equal width, as shown.

- (i) Find the value of $\int_1^2 \frac{dx}{x^2}$ 1
- (ii) Show that the area of the shaded rectangle is $\frac{n}{(n+1)^2}$ 3
- (iii) Find an expression for the area of the rectangle directly to the right of the shaded one. 1
- (iv) You are given that the area of the last rectangle on the right is $\frac{n}{(2n)^2}$ 2
Show that

$$\lim_{n \rightarrow \infty} n \left(\frac{1}{(n+1)^2} + \frac{1}{(n+2)^2} + \dots + \frac{1}{(2n)^2} \right) = \frac{1}{2}$$

END OF EXAMINATION PAPER

SOLUTIONS

QUESTION 1

(a) (i) $y = x^2$
 $y = 3x - 2$

$$x^2 = 3x - 2$$

$$\therefore (x-2)(x-1) = 0$$

$$\therefore \begin{cases} x=2 \\ y=4 \end{cases} \text{ or } \begin{cases} x=1 \\ y=1 \end{cases} \quad (1)$$

(ii) $A = \int_1^2 3x - 2 - x^2 dx \leftarrow (1)$

$$= \left[\frac{3}{2}x^2 - 2x - \frac{1}{3}x^3 \right]_1^2$$

$$= (6 - 4 - \frac{8}{3}) - (\frac{3}{2} - 2 - \frac{1}{3})$$

$$= \frac{1}{6} \text{ u}^2 \leftarrow (1)$$

OR, 1 MARK for each area

(b) $\frac{dy}{dx} = 6x^2 - 2x + k$

At $x=0$, $\frac{dy}{dx} = 0$

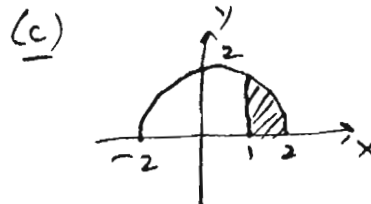
$$\therefore k = 0$$

$$\therefore \frac{dy}{dx} = 6x^2 - 2x \quad (1)$$

$$y = 2x^3 - x^2 + c$$

At $x=0$, $y=4 \therefore c=4$

$$\therefore y = 2x^3 - x^2 + 4 \quad (1)$$



BOWL /
CONTACT LENS.

(1)

(ii) $VOL = \pi \int_1^2 y^2 dx \quad (1)$

$$= \pi \int_1^2 (4 - x^2) dx \quad (1)$$

$$= \pi \left[4x - \frac{1}{3}x^3 \right]_1^2$$

$$= 5\pi/3 \text{ cu units} \quad (1)$$

(d) Let $u = 1 - x^2 \Rightarrow \frac{du}{dx} = -2x$

$$\therefore \int x\sqrt{1-x^2} dx \quad \frac{du}{dx} = -\frac{du}{2x}$$

$$= \int x\sqrt{u} \left(-\frac{du}{2x}\right) \quad (1)$$

$$= -\frac{1}{2} \int \sqrt{u} du \leftarrow$$

$$= -\frac{1}{2} \cdot \frac{2}{3} u^{3/2} + k$$

$$= -\frac{1}{3} (1-x^2)^{3/2} + k \quad (1)$$

OR,

$$\int x\sqrt{1-x^2} dx$$

$$= \frac{2}{3} (1-x^2)^{3/2} \cdot \frac{1}{2} + k$$

$$= -\frac{1}{3} (1-x^2)^{3/2} + k.$$

(2)

[guess and correct
method]

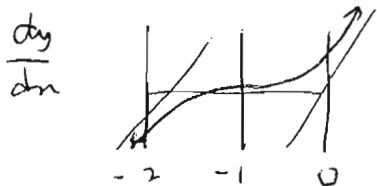
QUESTION 2:

(a) $y = (1-x)(1+x)^3$

$$\begin{aligned}\frac{dy}{dx} &= (1+x)^3(-1) + (1-x)3(1+x)^2 \\ &= (1+x)^2 [-1-x+3-3x] \quad \textcircled{1} \\ &= (1+x)^2 (2-4x) \\ &= 2(1+x)^2 (1-2x) \leftarrow \textcircled{1}\end{aligned}$$

(b) At S.P.'s $\frac{dy}{dx} = 0$

$$\begin{aligned}\textcircled{1} \rightarrow \begin{cases} x = -1 \\ y = 0 \\ y'' = 0 \end{cases} \quad \text{or} \quad \begin{cases} x = \frac{1}{2} \quad \textcircled{1} \\ y = \frac{27}{16} \\ y'' < 0 \Rightarrow \text{max.} \quad \textcircled{1} \end{cases} \\ \text{at } (\frac{1}{2}, \frac{27}{16})\end{aligned}$$



I. P. at $(-1, 0)$ $\textcircled{1}$

(c) At I. P.'s $\frac{d^2y}{dx^2} = 0$

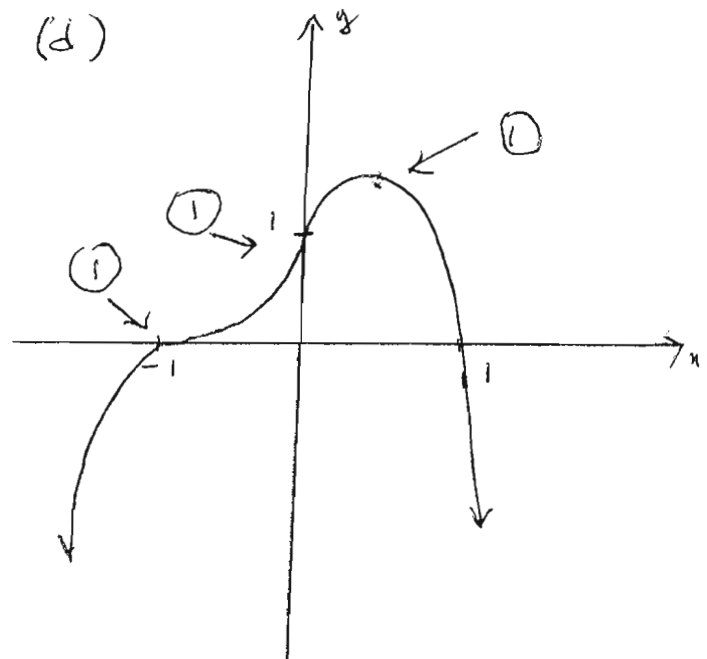
$$\therefore -12x(x+1) = 0$$

$$\textcircled{1} \begin{cases} x = 0 & \text{or} & x = -1 \\ y = 1 & \text{found} \end{cases}$$

$$y'' \begin{array}{|c|c|c|} \hline -\frac{1}{2} & 0 & \frac{1}{2} \\ \hline + & 0 & - \\ \hline \end{array} \leftarrow \textcircled{1}$$

$\therefore$ I. P.'s at $(0, 1)$ and $(-1, 0)$

(d)



QUESTION 3:

(c) (i) $\int x + \frac{1}{x} dx = \frac{1}{2}x^2 + -\frac{1}{x} + k.$
(2) $\begin{cases} \frac{1}{2}x^2 + -\frac{1}{x} + k \end{cases}$

Penalty of ① for
no constant here
only

(ii) $\sqrt{2x^{\frac{1}{2}} \cdot 4} = 4$
 ↑ ↑
 ① ①

(b) (i) $A_R = 3x^2$ $A_S = \left(\frac{28-8x}{4}\right)^2 \leftarrow ①$
 ↑
 ①

$\therefore A = 3x^2 + 49 - 28x + 4x^2$
 $= 7x^2 - 28x + 49.$

(ii) $\frac{dA}{dx} = 14x - 28$ $\frac{d^2A}{dx^2} = 14 > 0 \Rightarrow \text{min.}$ $\leftarrow ①$

$\therefore x = 2 \leftarrow ①$

$\therefore y = 3.$

$\therefore \text{Area of square} = 9 \text{ m}^2. \leftarrow ①$

(c) $\frac{d}{dx} (1-2x)^{\frac{1}{2}} = \frac{1}{2}(-2)(1-2x)^{-\frac{1}{2}}.$
 $= -\frac{1}{\sqrt{1-2x}} \quad ①$

$\therefore \int \frac{dx}{\sqrt{1-2x}} = -\sqrt{1-2x} + k \quad ①$ for minus
no penalty for no k.

Question 4:

$$(a) \quad A_1 = \frac{1}{3} \cdot 0.5 [1 + 6.596 + 2.718] \leftarrow (1) \quad (1.719)$$

$$A_2 = \frac{1}{3} \cdot 0.5 [2.718 + 17.928 + 7.385] \leftarrow (1) \quad (4.6725)$$

$$\therefore \int f(x) dx \approx 6.39$$

[note: there are many ways to do this!]

$$(b) \quad \sin 2x = \sqrt{3}/2 \quad 0 \leq 2x \leq 2\pi$$

$$\therefore 2x = \pi/3, 2\pi/3, \dots \leftarrow (1) \text{ or equivalent}$$

$$\therefore x = \pi/6, \pi/3 \leftarrow (2)$$

$$(c) \quad u = 1-x \Rightarrow du/dx = -1 \quad \left\{ \begin{array}{l} x=0 \quad u=1 \\ x=1 \quad u=0 \end{array} \right\} \leftarrow (1)$$

$$\therefore dx = -du$$

$$\therefore \int_0^1 x\sqrt{1-x} dx = -\int_1^0 (1-u)\sqrt{u} du \leftarrow (1)$$

$$= \int_0^1 -\sqrt{u} + u\sqrt{u} du \leftarrow (1)$$

$$= \left[-\frac{2}{3}u^{3/2} \right]_0^1 + \left[\frac{2}{5}u^{5/2} \right]_0^1$$

$$= \frac{2}{3} + \left(-\frac{2}{5} \right) \leftarrow (2)$$

$$= \frac{4}{15}$$

OR //

$$u = 1-x \Rightarrow dx = -du$$

$$\therefore \int x\sqrt{1-x} = \int (1-u)\sqrt{u} (-du) \leftarrow (1)$$

$$= u\sqrt{u} - \sqrt{u} + k$$

$$= \frac{2}{5}u^{5/2} - \frac{2}{3}u^{3/2} + k \leftarrow (1)$$

$$= \frac{2}{5}(1-x)^{5/2} - \frac{2}{3}(1-x)^{3/2} \leftarrow (1)$$

$$\therefore \text{Def. int} = \left[\frac{2}{5}(1-x)^{5/2} - \frac{2}{3}(1-x)^{3/2} \right]_0^1 \leftarrow (1)$$

$$= 0 - \frac{2}{5} + \frac{2}{3}$$

$$= \frac{4}{15}$$

$\leftarrow (1)$

Question 5:

$$\underline{a(i)} \quad \cos \theta - \sin \theta = \sqrt{2} (\cos \theta \frac{1}{\sqrt{2}} - \sin \theta \frac{1}{\sqrt{2}}) \\ = \sqrt{2} \cos (\theta + \alpha)$$

$$\text{where } \cos \alpha = \frac{1}{\sqrt{2}}$$

$$\therefore \alpha = \frac{\pi}{4}$$

$$\therefore \cos \theta - \sin \theta = \sqrt{2} \cos (\theta + \frac{\pi}{4})$$

$$\uparrow \textcircled{1} \quad \uparrow \textcircled{1}$$

$$\underline{(ii)} \quad \cos \theta - \sin \theta = 1 \Rightarrow \sqrt{2} \cos (\theta + \frac{\pi}{4}) = 1$$

$$\therefore \cos (\theta + \frac{\pi}{4}) = \frac{1}{\sqrt{2}}$$

$$\therefore \theta + \frac{\pi}{4} = \frac{\pi}{4} \text{ or } \frac{7\pi}{4}$$

$$\therefore \theta = 0 \text{ or } \theta = \frac{3\pi}{2} \leftarrow \textcircled{1} \text{ for each.}$$

$$\underline{(iii)} \quad \theta = 0 \text{ or } \theta = -\frac{\pi}{2} \quad \textcircled{1}$$

$$\underline{(b)} \quad \underline{(i)} \quad \text{Since } \angle OP = 90^\circ, \quad TP^2 = 4a^2 - a^2 \\ TP = a\sqrt{3} \leftarrow \textcircled{1}$$

$$\therefore \text{Area } \triangle OTP = \frac{1}{2} \cdot 0 \cdot a\sqrt{3} \\ = \frac{a^2\sqrt{3}}{2} \leftarrow \textcircled{1}$$

$$\underline{(ii)} \quad \cos \theta = \frac{a}{2a} = \frac{1}{2}$$

$$\therefore \theta = \frac{\pi}{3}$$

$$\therefore \angle TOR = 2\frac{\pi}{3} \leftarrow \textcircled{1}$$

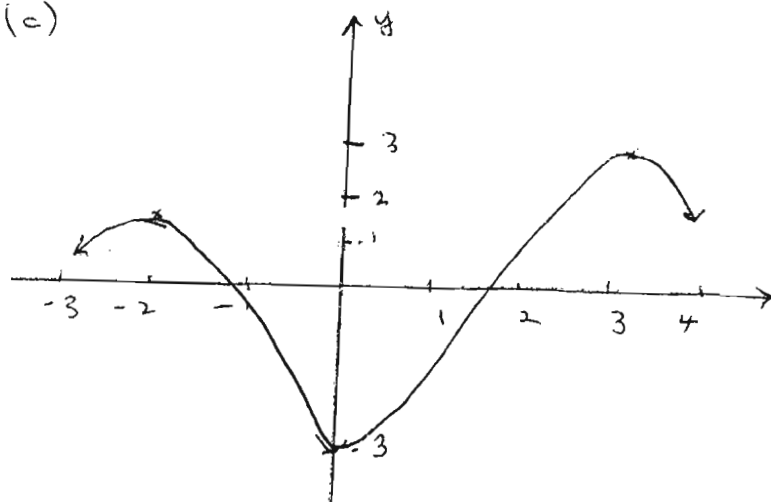
$$\underline{(iii)} \quad \text{Area } \triangle = 2 \times \frac{a^2\sqrt{3}}{2} \leftarrow \textcircled{1} \\ = a^2\sqrt{3}$$

$$\text{Area sector } OTR = \frac{1}{2} r^2 \theta \\ = \frac{1}{2} (a^2) \cdot 2\frac{\pi}{3} \\ = \frac{\pi a^2}{3} \leftarrow \textcircled{1}$$

$$\therefore \text{Area shaded} = a^2 \left(\sqrt{3} - \frac{\pi}{3} \right) \leftarrow \textcircled{1} \\ = \frac{a^2}{3} (3\sqrt{3} - \pi)$$

QUESTION 6:

(c)



③ 1 each for each of the T.P.'s

① for showing the points $(-2, 1)$ $(0, 0)$ $(1, -3)$ $(3, 3)$

(b)

$$(i) \int_1^2 \frac{dx}{x^2} = -x^{-1} \Big|_1^2 = \frac{1}{2} \quad \text{①}$$

$$(ii) \text{ width} = \frac{1}{n}, \text{ height is } \frac{1}{(1+\frac{1}{n})^2} \therefore \text{Area} = \frac{1}{n} \left[\frac{1}{(1+\frac{1}{n})^2} \right] \\ \uparrow \text{①} \quad \uparrow \text{①} \\ = \frac{1}{n} \cdot \frac{n^2}{(n+1)^2} \\ = \frac{n}{(n+1)^2} \quad \text{①}$$

$$(iii) A_n = \frac{n}{(n+1)^2} \quad \text{①}$$

(iv) Sum of areas of rectangles

$$= \frac{n}{(n+1)^2} + \frac{n}{(n+2)^2} + \dots + \frac{n}{(2n)^2} \\ = n \left[\frac{1}{(n+1)^2} + \frac{1}{(n+2)^2} + \dots + \frac{1}{(2n)^2} \right] \quad \text{①}$$

The exact area is the limit of this sum as $n \rightarrow \infty$

and from part (i) this exact area is $\frac{1}{2}$

$$\therefore \lim_{n \rightarrow \infty} n \left[\frac{1}{(n+1)^2} + \frac{1}{(n+2)^2} + \dots + \frac{1}{(2n)^2} \right] = \frac{1}{2} \quad \text{①}$$